Push Factors and Filipino Emigration: A Critical Review of Agbola and Acupan (2010)

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Abstract

This review assesses Agbola and Acupan's (2010) study of the push factors behind emigration from the Philippines. The authors estimate an error correction model on yearly national data for 1975–2005 and find that unemployment raises emigration, while adult literacy, political instability and population density lower it. The methods are applied carefully, but the sample is small (31 observations for ten explanatory terms), two proxies are weak, and pull factors in destination countries are left out. We suggest four extensions, the most important being separate models for temporary and permanent migration.

Keywords: international migration, push factors, error correction model, Philippines

JEL codes: F22, J61, O15

1 Introduction

About 8.2 million Filipinos live or work in more than 193 countries, and they sent home US$14.4 billion in remittances in 2007 (Agbola and Acupan 2010). Sending workers abroad began in the 1970s as a short-term fix for unemployment. It has since become a permanent part of the economy. Yet, as the authors note, the causes of emigration from the Philippines are "not well established."

The study asks one question: how do economic, demographic and political factors at home affect emigration? A *push factor* is a condition at home that pushes people out, such as high unemployment. A *pull factor* is a condition abroad that draws them in, such as higher wages.

[1] Graduate student in economics. This document is an example of the Lian signature style; the review text is a short illustration.

2 Theoretical framework

The authors combine four theories. In the classic models, people migrate when they expect a better life abroad (Ravenstein 1885; Todaro 1969). Human capital theory treats migration as an investment, so more educated people should migrate more. The Roy model, as applied by Borjas (1987), says that income inequality decides which skill groups leave.

The resulting model, in words, is

$$m_t = f\left(c_t, \ell_t, g_t, g_t^2, s_t, \pi_t, d_t, u_t, y_t\right) \quad (1)$$

where m_t is the migration share in year t, c_t the cost of migration, ℓ_t adult literacy, g_t income inequality, s_t political instability, π_t inflation, d_t population density, u_t unemployment and y_t per capita income.

3 Results

Table 1 reports the main estimates. Four push factors are statistically significant at the 5% level, and the model explains 63% of the year-to-year changes in the migration share.

Table 1. Error correction model estimates of the migration share, Philippines, 1975–2005

Explanatory variable	Coefficient	*t*-statistic
Unemployment	0.236***	(4.629)
Adult literacy	−0.659***	(−5.225)
Political instability	−0.482***	(−3.444)
Population density	−0.254**	(−2.756)
Income inequality	−171.195	(−1.594)
Income inequality²	165.927	(1.323)
Inflation	−0.036	(−1.327)
Cost of migration	−0.002	(−0.429)
Per capita income	−0.0002	(−0.293)
Adjusted R^2	0.63	

Notes: *t*-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The error correction term is −0.776, so about 78% of any gap from the long-run relationship closes within a year.

Source: Agbola and Acupan (2010), Table 4.

The positive unemployment coefficient fits the idea of emigration as a "safety valve" when there are not enough jobs at home. The negative literacy coefficient is the opposite of what human capital theory predicts.

4 Critical assessment

The study's strengths are an important question and careful time-series steps: unit root tests, cointegration tests, a general-to-specific model and diagnostic tests that all pass. Its weaknesses are three. First, the sample is small: 31 yearly observations for ten explanatory terms. Second, airfare stands in for the full cost of migration, and adult literacy stands in for education.[2] Third, only push factors are modelled.

References

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[2] A *proxy* is a stand-in measure for something that cannot be measured directly.